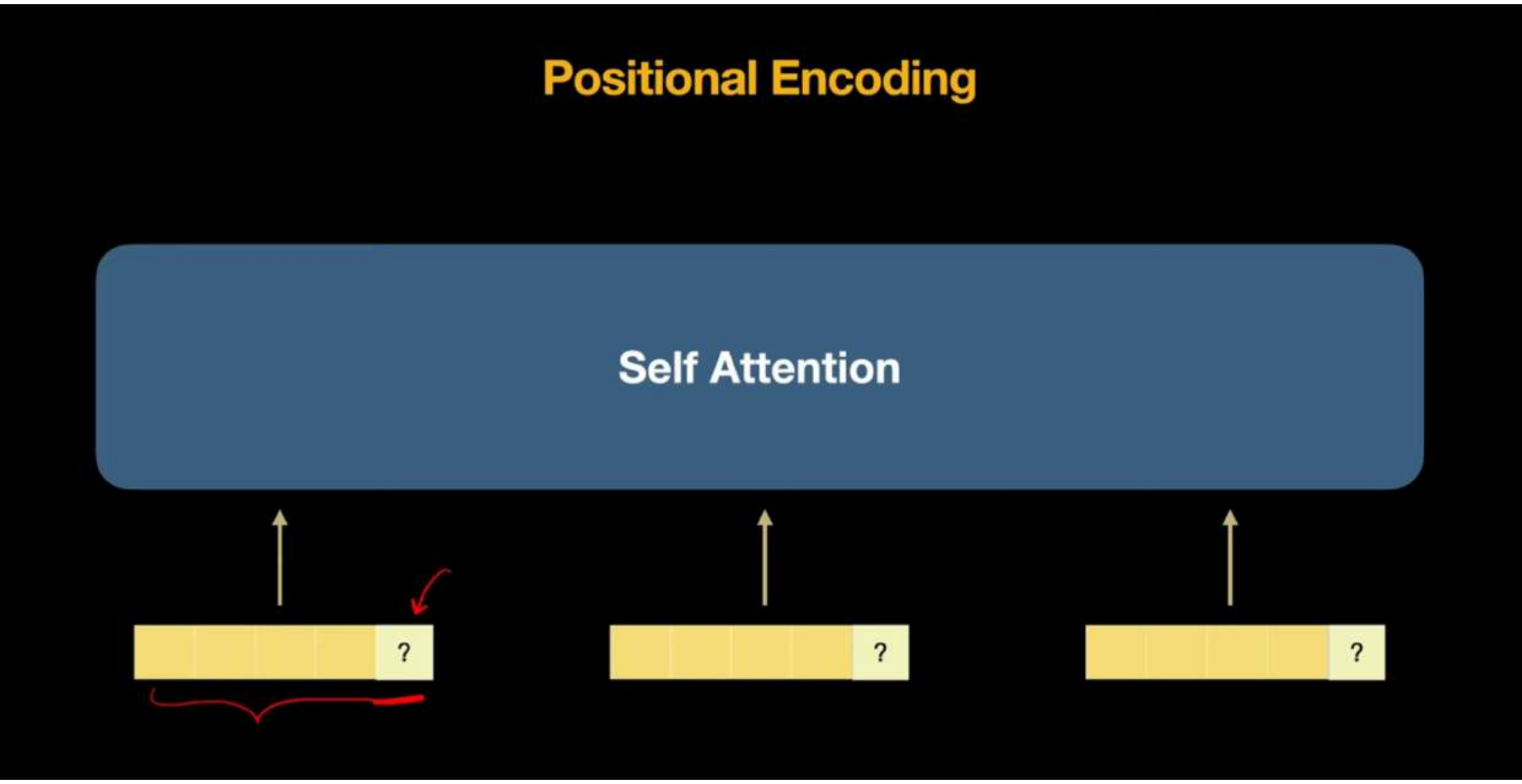

Positional Encoding
Self Attention
?
?
?

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
y-axis
0 1 2 3 4 5 6
y = sin(x)
y = cos(x)
y = sin(x/2)
y = cos(x/2)
d
512 dims
264 pairs of sine cosine
Sin(1) Cos(1) Sin(1/2) Cos(1/2)

dim. of position Encoding = 512

0

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

pos = 1	$\sin(1)$	$\cos(1)$	$\sin\left(\frac{1}{2.5}\right)$	$\cos\left(\frac{1}{2.5}\right)$	$\sin\left(\frac{1}{6.3}\right)$	$\cos\left(\frac{1}{6.3}\right)$	. .	. .

pos 2

Dimension $i = 0$ (Lowest dimension)

$$\text{Scaling factor} = 10{,}000^{\frac{2\cdot 0}{512}} = 10{,}000^{0} = 1$$

$$\text{Argument for sine} = \frac{1}{1} = 1$$

Dimension $i = 1$

$$\text{Scaling factor} = 10{,}000^{\frac{2\cdot 1}{512}} = 10{,}000^{0.00390625} \approx 2.511$$

$$\text{Argument for sine} = \frac{1}{2.511} \approx 0.398$$

Dimension $i = 2$

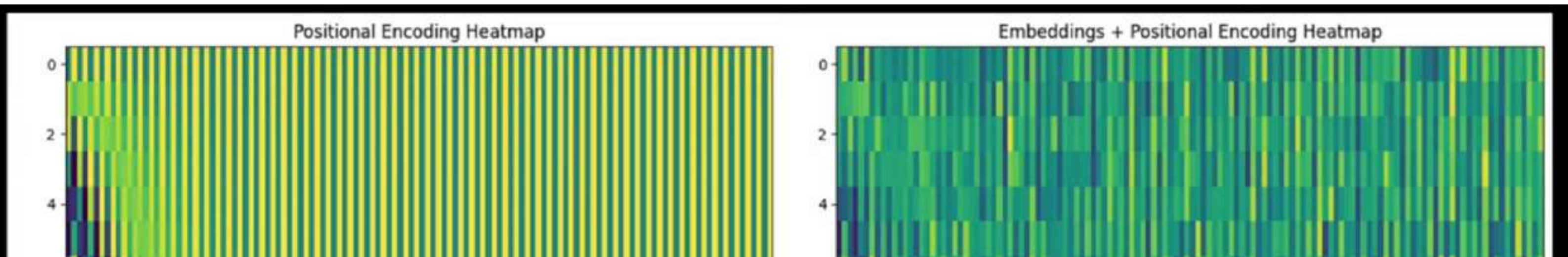
$$\text{Scaling factor} = 10{,}000^{\frac{2\cdot 2}{512}} = 10{,}000^{0.0078125} \approx 6.31$$

$$\text{Argument for sine} = \frac{1}{6.31} \approx 0.159$$

Dimension $i = 3$

$$\text{Scaling factor} = 10{,}000^{\frac{2\cdot 3}{512}} = 10{,}000^{0.01171875} \approx 15.85$$

$$\text{Argument fo.ne} = \frac{1}{15.85} \approx 0.063$$



Positional Encoding Heatmap
Position Index
Embedding Dimension
0
2
4
6
8
10
12
14
0
20
40
60
80
100
120
Embeddings + Positional Encoding Heatmap
Word Index
Embedding Dimension